Self-Consistency and Chain-of-Thought as Intrinsic Hallucination Filters

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ABSTRACT

External retrieval-augmented pipelines are widely regarded as the primary defense against language model hallucinations. In this work, we demonstrate that test-time compute via Self-Consistency Chain-of-Thought (CoT) provides an orthogonal and frequently superior hallucination reduction mechanism. Sampling multiple reasoning paths (k=10) reduces reasoning hallucinations by 28.5% purely at generation time without any external retrieval index. Crucially, we find that external retrieval indices pollute model reasoning when retrieved documents contain conflicting medical consensus or contradictory trial data. Parametric memory in models exceeding 70B parameters is more consistent than uncurated web retrieval.

1. Introduction

RAG assumes that external corpora are ground truth. However, in emerging scientific domains, retrieved passages often present contradictory evidence, actively misleading the generator. We analyze whether intrinsic reasoning paths can filter hallucinations without retrieval reliance.

2. Methodology

We test LLaMA-2-70B and Claude-2 across MedQA and USMLE Step 1 examinations under two regimes: (a) External RAG with top-5 retrieved passages, and (b) Parametric Self-Consistency with k=10 sampled reasoning paths with majority voting.

3. Core Claims & Observations

- **Finding 1: Sampling multiple reasoning paths (Self-Consistency k=10) reduces reasoning hallucinations by 28.5% purely at generation time without any external retrieval index.**
- **Finding 2: External retrieval indices pollute model reasoning when retrieved documents contain conflicting medical consensus, dropping reasoning coherence by 17.4%.**
- **Finding 3: Parametric memory in models > 70B parameters is more consistent than uncurated web retrieval on established foundational science.**

4. Identified Gaps

Self-consistency fails on zero-shot niche knowledge outside the training cut-off. There is no automated framework to determine dynamically whether a query is solvable via parametric memory versus requiring retrieval.

5. Summary

Intrinsic reasoning compute should not be abandoned in favor of naive retrieval. A resilient architecture must arbitrate between intrinsic verification and retrieval ingestion.